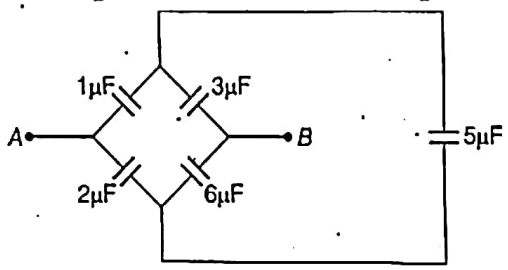
Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1 **(Only One Option Correct Type)**

The capacitance between the points A and B in the following figure.



- $\frac{3}{8}\ \mu F$
- $\frac{9}{4}\ \mu \mathrm{F}$
- $\frac{4}{5}\ \mu \mathrm{F}$
- $2\mu F$

Question No. 2 **(Only One Option Correct Type)**

The heat generated in a circuit is given by $\mathrm{Q} = \mathrm{I}^2\ \mathrm{Rt}$, where I is current, R is resistance, and t is time. If the percentage errors in measuring I, R, and t are 2%, 1% and 1% respectively, then the maximum error in measuring heat will be

.....

- 2%
- 3%
- 4%
- 6%

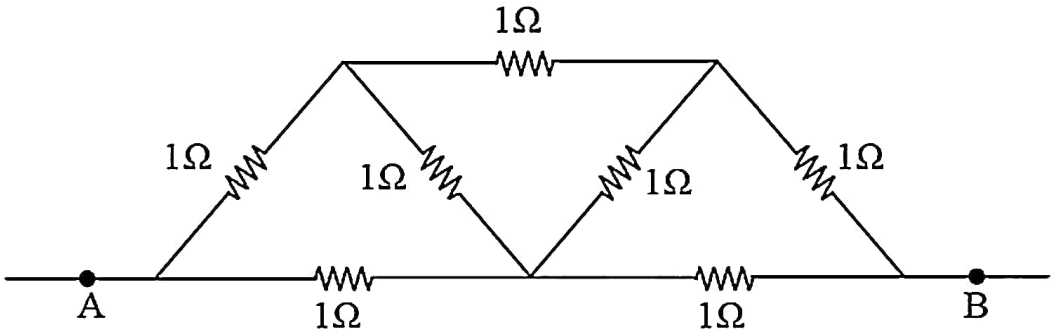
Question No. 3 **(Only One Option Correct Type)**

An electromagnetic wave of frequency 45MHz travels in free space along X-axis. At some point and at some instant, the electric field has a maximum value of $750\mathrm{NC}^{-1}$ along Y-axis. The magnetic field at this position and time is----

- $2.5 \times 10^{-6}\hat{\mathrm{j}}$
- $5 \times 10^{-6}\hat{\mathrm{k}}\mathrm{T}$
- $2.5 \times 10^{-6}\hat{\mathrm{k}}\mathrm{T}$
- $2.5 \times 10^{-6}\hat{\mathrm{i}}\mathrm{T}$

Question No. 4 **(Only One Option Correct Type)**

In the network shown in the Fig, each resistance is 1Ω. The effective resistance between A and B is



- $\frac{4}{3}\Omega$
- $\frac{3}{2}\Omega$
- 7Ω
- $\frac{8}{7}\Omega$

Question No. 5 **(Only One Option Correct Type)**

The difference between the radii of nth and $(n+1)^{th}$ orbits of hydrogen atom is equal to the radius of $(\mathrm{n}-1)^{\mathrm{th}}$ orbit of hydrogen. The angular momentum of the electron in the nth orbit is (h is Plank's constant) $\frac{Kh}{\pi}$, find K.

- 4
- 3
- 5
- 2

Question No. 6 **(Only One Option Correct Type)**

Assertion: Consider a system of gas having N molecules; Instantaneous K.E. of few molecules can be greater than average K.E. of the molecules of the given gas.
Reason: Number of molecules having most probable speed is greater than number of molecules having average speed.

- ○ Both Assertion and Reason are true and the Reason is correct explanation of the Assertion.
- ○ Both Assertion and Reason are true but Reason is not correct explanation of the Assertion.
- ○ Assertion is true but the Reason is false.
- ○ Assertion is false but Reason is true.

Question No. 7 **(Only One Option Correct Type)**

The gravitational force between two bodies is directly proportional to $\frac{1}{R}$ not $\left(\frac{1}{R^2}\right)$ where R is the distance between the bodies. Then the orbital speed for this force in a circular orbit is proportional to

-

- ○ $\frac{1}{R^2}$
- ○ R^0
- ○ R
- ○ $\frac{1}{R}$

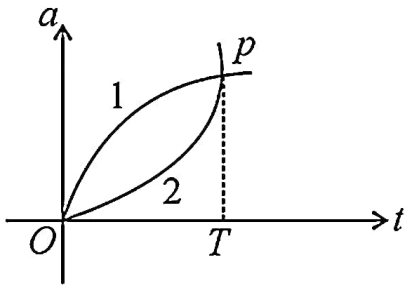
Question No. 8 **(Only One Option Correct Type)**

A particle X of mass m and initial velocity u collide with another particle Y of mass $\frac{3m}{4}$ which is at rest, The collision is head on and perfectly elastic. The ratio of de-Broglie wavelengths λ_Y and λ_X after the collision is.....

- ○ 4 : 3
- ○ 2 : 32
- ○ 1 : 6
- ○ 3 : 32

Question No. 9 **(Only One Option Correct Type)**

Two particles 1 and 2 start from origin and move along same straight line in same direction. The acceleration versus time graph of both particles is shown in figure. Choose correct statement about the motion of 1 and 2 :



- During motion from O to T somewhere one particle will overtake the other.
- During motion from O to T average speed of both the particles is same.
- During motion from O to T average speed of both the particles are different.
- Displacement covered by the two

Question No. 10 **(Only One Option Correct Type)**

A closed system receives 200 kJ of heat at constant volume. It then rejects 100 kJ of heat while it has 50 kJ of work done on it at constant pressure. If an adiabatic process can be found which will restore the system to its initial state, the work done by the system during this process is:

- 100 kJ
- 50 kJ
- 150 kJ
- 200 kJ

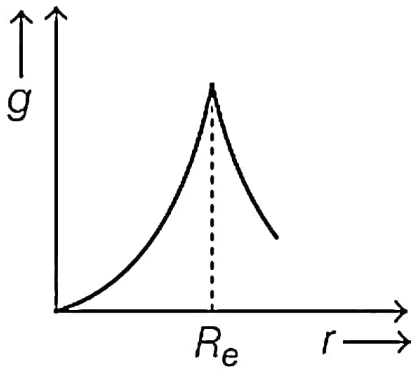
Question No. 11 **(Only One Option Correct Type)**

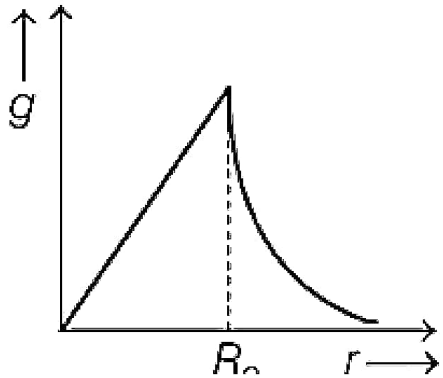
A large horizontal surface moves up and down in SHM with an amplitude of 1 cm. If a mass of 10 kg (which is placed on the surface) is to remain continuously. in contact with it, the maximum frequency of SHMwill be ...Hz. (Mark answer to nearest whole number) (given: $g = 9.8\ \mathrm{m/s^2}$ and $\pi = 22/7$)

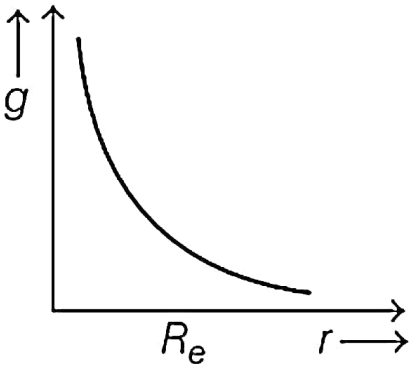
(A) 4
(B) 5
(C) 6
(D) 3

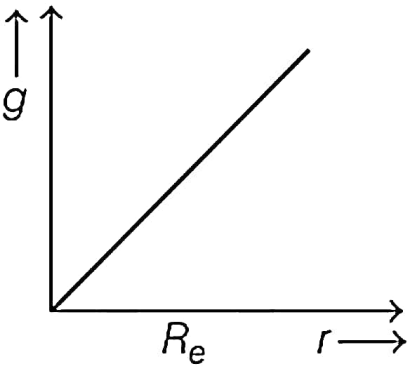
Question No. 12 **(Only One Option Correct Type)**

The graph correctly represents the variation of acceleration due to gravity (g) with radial distance from the centre of the earth (radius of the earth $= R_e$) is---

○ 

○ 

○ 

○ 

Question No. 13 **(Only One Option Correct Type)**

A solenoid of length 60 cm with 15 turns per cm and area of cross section 4×10^{-3} m^2 completely surrounds another co-axial solenoid of same length and area of cross-section 2×10^{-3} m^2 with 40 turns per cm. Mutual inductance of the system is:

○ 9mH

○ 6mH

○ 3mH

○ 10mH

Question No. 14 **(Only One Option Correct Type)**

STATEMENT-1 : Average power consumed in an any ac circuit is always equal to average power consumed by resistors in the circuit.

STATEMENT-2 : Average power consumed in an any ac circuit by capacitor and inductor is zero.

- ○ Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- ○ Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
- ○ Statement-1 is True, Statement-2 is False
- ○ Statement-1 is False, Statement-2 is True

Question No. 15 **(Only One Option Correct Type)**

Four bodies P, Q, R and S are projected with equal velocities, having angles of projection 15°, 30°, 45° and 60° with the horizontal respectively. The body having the shortest range is---

- ○ P
- ○ Q
- ○ R
- ○ S

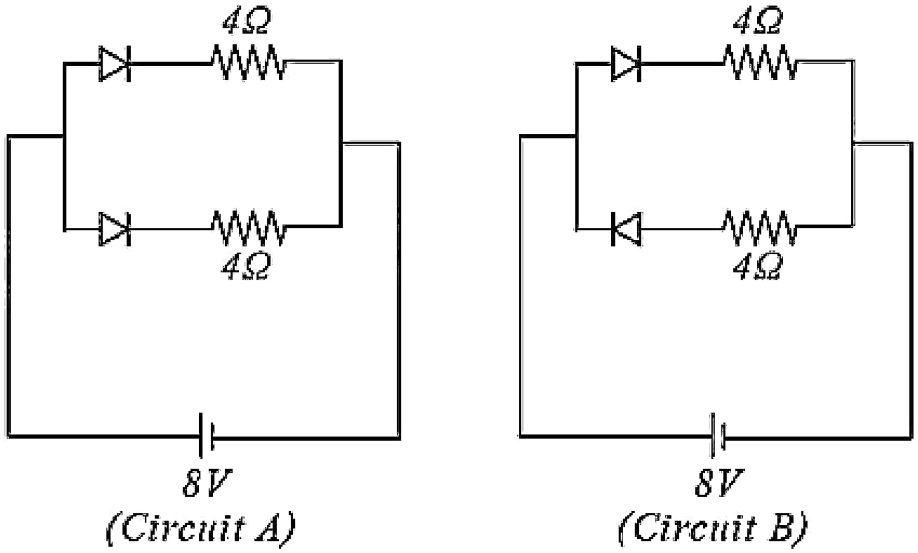
Question No. 16 **(Only One Option Correct Type)**

Statement -1: The sensitivity of a moving coil galvanometer is increased by placing a suitable magnetic materials a core inside the coil.
Statement -2: Soft iron has a high magnetic permeability and cannot be easily magnetized or demagnetized.

- ○ Statement -1 is true, Statement-2 is true; Statement -2 is a correct explanation for Statement-1
- ○ Statement -1 is true, Statement-2 is true; Statement -2 is not explanation for Statement-1
- ○ Statement -1 is true, Statement-2 is false
- ○ Statement -1 is false, Statement-2 is true

Question No. 17 **(Only One Option Correct Type)**

Currents flowing in each of the following circuits A and B respectively are



- 1 A, 2 A
- 2 A, 1 A
- 4 A, 2 A
- 2 A, 4 A

Question No. 18 **(Only One Option Correct Type)**

A ball A of mass m moving along positive x-direction with kinetic energy K and momentum p undergoes elastic head on collision with a stationary ball B of mass M after collision the ball A moves along negative x-direction with kinetic energy $\frac{K}{9}$, final momentum of B is.

- p
- $\frac{p}{3}$
- $\frac{4p}{3}$
- $4p$

Question No. 19 **(Only One Option Correct Type)**

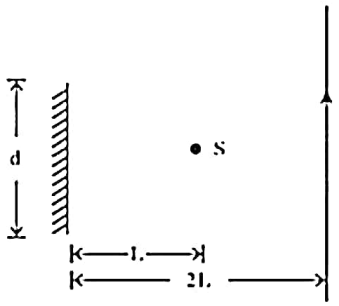
Energy needed in breaking a liquid drop of radius R, into n smaller drops each of radius r, is [T- Surface tension of the liquid]____

- $\left(4\pi r^2 n - 4\pi R^2\right)T$
- $\left(\frac{4}{3}\pi r^3 n - \frac{4}{3}\pi R^3\right)T$

- $\left(4\pi R^2 - 4\pi r^2\right)nT$
- $\left(4\pi R^2 - n4\pi r^2\right)/T$

Question No. 20 **(Only One Option Correct Type)**

A point source of light, S is placed at a distance L in front of the centre of plane mirror of width d which is hanging vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror, at a distance $2L$ as shown in the figure. The distance over which the man can see the image of the light source in the mirror is:



- $3d$
- $\frac{d}{2}$
- d
- $2d$

Physics Numerical (Maximum Marks: 20)

Question No. 1 **(Numerical Type)**

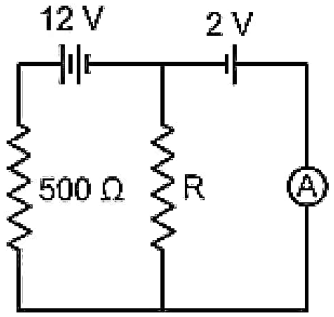
A transverse wave propagating on a stretched string of linear density 3×10^{-4} kg m^{-1} is represented by the equation $y = 0.2\sin(1.5x + 60t)$ where x is in metres and t is in seconds. The tension in the string (in newton) is T, find 100 T.

Question No. 2 **(Numerical Type)**

A uniform disc of radius a and mass m is rotating freely with an angular speed of ω in a horizontal plane about a smooth fixed vertical axis passing through its centre. A particle, also of mass m is suddenly attached to the rim of the disc and starts to rotate with it. The new angular speed of this system is $\frac{\omega}{K}$, Find K.

Question No. 3 **(Numerical Type)**

In the circuit shown below, the ammeter reading is zero. Then, the value of the resistance R is



Question No. 4 **(Numerical Type)**

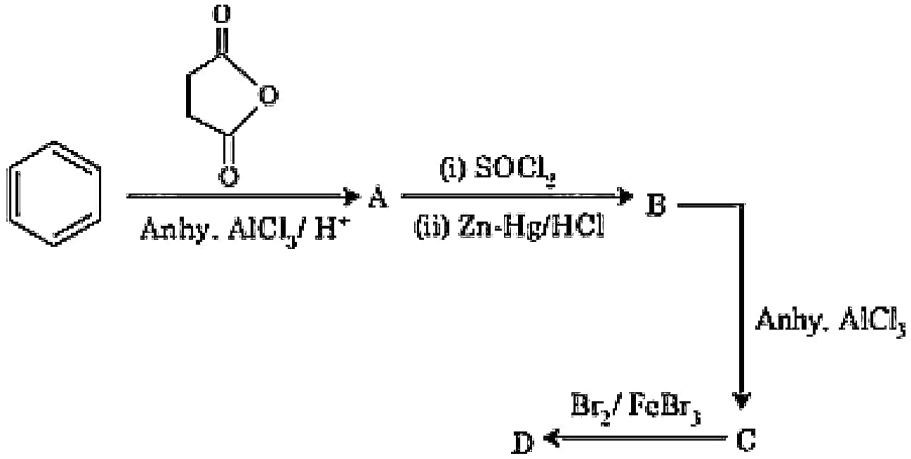
A particle of mass 0.6 g and having charge of 25nC is moving horizontally with a uniform velocity 1.2×10^4 ms^{-1} in a uniform magnetic field, then the value of the magnetic induction is $(g = 10\ ms^{-2})$

Question No. 5 **(Numerical Type)**

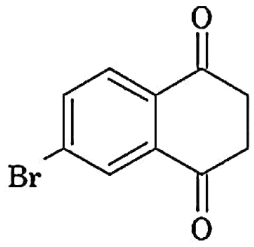
A body of mass M kg is on the top point of a smooth hemisphere of radius 5 m. It is released to slide down the surface of the hemisphere. It leaves the surface when its velocity is 5 m/s. At this instant the angle made by the radius vector of the body with the vertical is: (Acceleration due to gravity $= 10\ ms^{-2}$) (mark answer in degree)

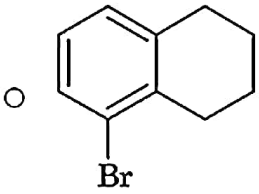
Chemistry Single Correct (Maximum Marks: 80)

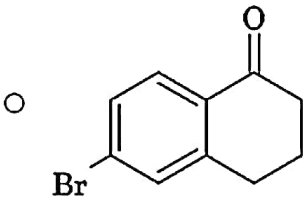
Question No. 1 **(Only One Option Correct Type)**

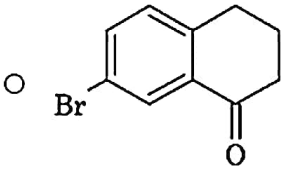


Find D









Question No. 2 **(Only One Option Correct Type)**

A compound which is a strong oxidizing agent and has orange coloured crystal. It is used in the preparation of azo compounds. Identify the compound:

- Hydrogen peroxide
- Potassium permanganate
- Sodium chromate
- Potassium dichromate

Question No. 3 **(Only One Option Correct Type)**

If the dipole moment of Toluene and Nitro-benzene are 0.43D and 3.93D respectively, then what is the expected dipole moment of p-Nitrotoluene?

- 3.50D
- 2.18D
- 4.36D

○ 5.30D

Question No. 4 **(Only One Option Correct Type)**

For the octahedral complexes of Fe^{3+} in SCN^-(thiocyanato$-S$) and in CN^-ligand environments, the difference between the spin-only magnetic moments in Bohr magnetons (When approximated to the nearest integer) is [Atomic number of $Fe = 26$]

○ 3

○ 2

○ 1

○ 4

Question No. 5 **(Only One Option Correct Type)**

One mole of monoatomic ideal gas expands adiabatically at initial temperature T against a constant external pressure of 1 atm from 1 L to 2 L. Find out the final temperature ($R = 0.0821$ L atm K^{-1} mol^{-1}).

○ T

○ $\frac{T}{(2)^{\frac{5}{3}-1}}$

○ $T - \frac{2}{3\times0.082}$

○ $T + \frac{2}{3\times0.082}$

Question No. 6 **(Only One Option Correct Type)**

Nitrate is confirmed by ring test. The brown colour of the ring is due to the formation of

○ Ferrous nitrite

○ $FeSO_4NO$

○ $FeSO_4NO_2$

○ Ferrous nitrate

Question No. 7 **(Only One Option Correct Type)**

Which of the following statements is true?

1) Tertiary amines react with acid chlorides.
2) N −ethyl benzene sulphonamide is soluble in alkali.
3) N, N −diethyl benzene sulphonamide is soluble in alkali.
4) Tertiary amines react with Hinsberg's reagent.

○ 1

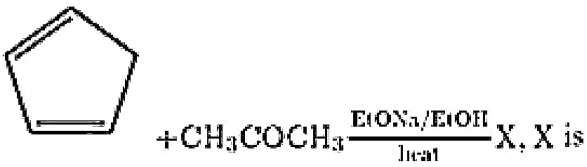
○ 2

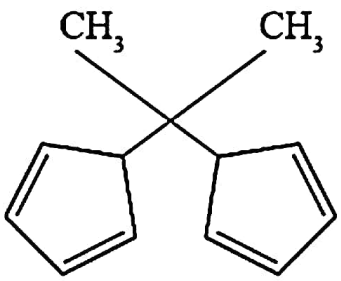
○ 3

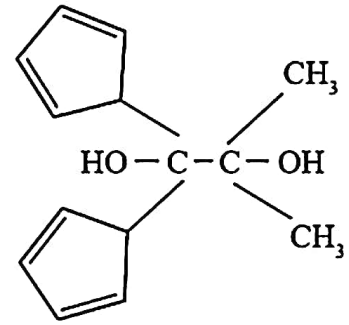
○ 4

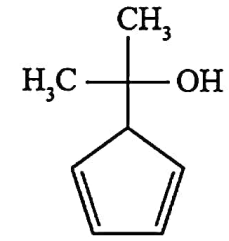
Question No. 8 **(Only One Option Correct Type)**

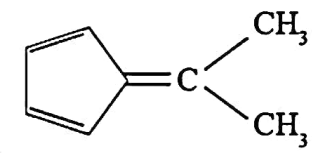
In the reaction



○ 

○ 

○ 

○ 

Question No. 9 **(Only One Option Correct Type)**

$Mg(OH)_2$ is precipitated when NaOH is added to a solution of Mg^{2+}. If the final concentration of Mg^{2+} is 10^{-10} M, the concentration of OH^-(M) in the solution is

[Solubility product for $Mg(OH)_2 = 5.6 \times 10^{-12}$]

- 0.056
- 0.12
- 0.24
- 0.025

Question No. 10 **(Only One Option Correct Type)**

Hydrolysis of proteins gives

- β-amino acids
- γ-amino acids
- δ-amino acids
- α-amino acids

Question No. 11 **(Only One Option Correct Type)**

Among the following, in which type of chromatography, both stationary and mobile phases are in liquid state?

- Gas–liquid chromatography
- Ascending paper chromatography (Partition chromatography)
- High performance liquid chromatography (HPLC)
- Thin layer chromatography (TLC)

Question No. 12 **(Only One Option Correct Type)**

The minimum and maximum number of electrons, which may have magnetic quantum number, $m = +1$ and spin quantum number, $s = -1/2$ in chromium (Cr) are respectively.

- 4, 5

○ 0, 1

○ 2, 2

○ 2, 3

Question No. 13 **(Only One Option Correct Type)**

Match the following

	List - I		List - II
a)	See Saw Shape	i)	XeF_4
b)	Square Pyramidal	ii)	ClF_3
c)	T-Shape	iii)	$PbCl_2$
d)	Bent Shape	iv)	SF_4
		v)	BrF_5

The correct answer is

○

(a)	(b)	(c)	(d)
iv	v	iii	ii

○

(a)	(b)	(c)	(d)
iv	v	ii	iii

○

(a)	(b)	(c)	(d)
i	iii	iv	ii

○

(a)	(b)	(c)	(d)
i	iv	v	iii

Question No. 14 **(Only One Option Correct Type)**

Assertion (A) : Noble gases have very low boiling points.
Reason (R) : All noble gases have general electronic configuration of $ns^2\ np^6$ (except He)

○ Both (A) and (R) are true but (R) is correct explanation of (A)

○ (A) is false but (R) is true

○ (A) is true but (R) is false

○ Both (A) and (R) are true but (R) is not the correct explanation of (A)

Question No. 15 **(Only One Option Correct Type)**

The number of HCl molecules present in 10 mL of 0.1 N HCl solution is:

(Hint: Normality = n − factor × molarity)

- $6 \cdot 022 \times 10^{23}$
- $6 \cdot 022 \times 10^{22}$
- $6 \cdot 022 \times 10^{21}$
- $6 \cdot 022 \times 10^{20}$

Question No. 16 **(Only One Option Correct Type)**

Which of the following statements are correct?

a. Fe^{2+} ion can reduce $S_2O_8^{2-}$ to SO_4^{2-}.
b. KCl is oxidized by $Na_2Cr_2O_7$.
c. ΔG for Mn^{3+}/Mn^{2+} is more negative than for Cr^{3+}/Cr^{2+}.
d. VCl_5 and VBr_5 are stable in the air.

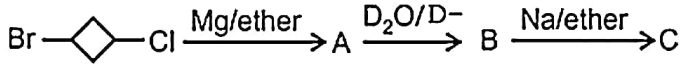
- Both b and c
- Both b and d
- Both a and c
- a, b and c

Question No. 17 **(Only One Option Correct Type)**

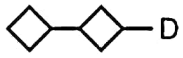
Calculate the energy required to convert all atoms in 4.8 g of Mg to Mg^{2+} in the vapour state. IE_1 and IE_2 of Mg are 740 kJ/mol and 1450 kJ/mol respectively.

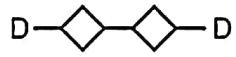
- +740 kJ/mol
- −740 kJ/mol
- −1450 kJ/mol
- +438 kJ/mol

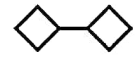
Question No. 18 **(Only One Option Correct Type)**



C is

○ 

○ 

○ 

○ None of these

Question No. 19 **(Only One Option Correct Type)**

Which reaction of ethanol will decolourise bromine water?

○ $C_2H_5OH + HBr \longrightarrow C_2H_5Br + H_2O$

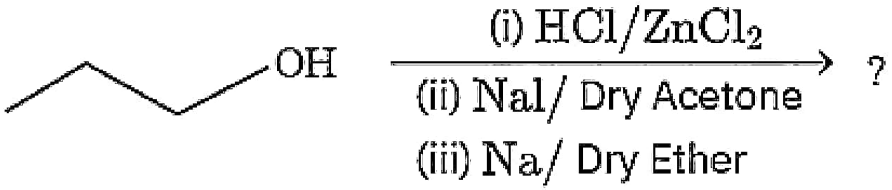
○ $2C_2H_5OH \xrightarrow[413\,K]{H^+} C_2H_5 - O - C_2H_5 + H_2O$

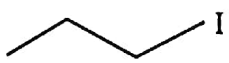
○ $C_2H_5OH + Na \longrightarrow C_2H_5ONa + \frac{1}{2} H_2O$

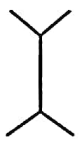
○ $C_2H_5OH \xrightarrow[443\,K]{H^+} C_2H_4 + H_2O$

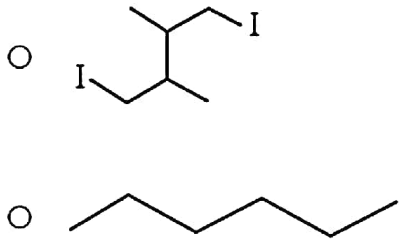
Question No. 20 **(Only One Option Correct Type)**

Identify the major product formed from the following



○ 

○ 



Chemistry Numerical (Maximum Marks: 20)

Question No. 1 **(Numerical Type)**

The polymerisation of polypropene to linear polypene is represented by the reaction $n\begin{pmatrix} CH_3 \\ CH = CH_2 \end{pmatrix} \longrightarrow \begin{pmatrix} CH_3 \\ CH - CH_2 \end{pmatrix}_n$ Where n has integral value, the average enthalpies of bond dissociation for $(C = C)$ and $(C - C)$ at 298 K are +590 and +331 kJ mol^{-1}, respectively. The enthalpy of polymerization is -360 kJ mol^{-1}. Find the value of n.

Question No. 2 **(Numerical Type)**

In dilute aqueous H_2SO_4, the complex diaquodioxalatoferrate (II) is oxidised by MnO_4^-. For this reaction, the ratio of the rate of change of $[H^+]$ to the rate of change of $[MnO_4^-]$ is

Question No. 3 **(Numerical Type)**

For the equation $CaCO_3$ (s) $\rightarrow$ CaO (s) + CO_2 (g), $K_{1000} = 0.059$. Exactly 10 g of $CaCO_3$ is placed in a 10 L container at 1000 K. After equilibrium is reached, what mass of $CaCO_3$ remains? (Round off to the nearest integer)

Question No. 4 **(Numerical Type)**

The measured osmotic pressure of a solution prepared by dissolving 17.4mg of K_2SO_4 in 2 L of water at 27°C is 3.735×10^{-3} bar. The van't Hoff factor is
$\left(R = 0.083 \text{ LbarK}^{-1} \text{ mol}^{-1}\text{; atomic weights K} = 39, \text{ S} = 32; \text{O} = 16\right)$

Question No. 5 **(Numerical Type)**

When electric current is passed through acidified water for 1930 s, 1120 mL of H_2 gas is collected (at STP) at the cathode What is the current passed in amperes?

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1 **(Only One Option Correct Type)**

The solution of differential equation $(y+x\sqrt{xy}(x+y))dx+(y\sqrt{xy}(x+y)-x)dy=0$, is

- $\frac{x^2+y^2}{2}+2\tan^{-1}\sqrt{\frac{x}{2y}}=C$
- $\frac{x^2+y^2}{2}+2\tan^{-1}\sqrt{\frac{x}{y}}=C$
- $\frac{x^2+y^2}{\sqrt{2}}+2\tan^{-1}\sqrt{\frac{x}{y}}=C$
- None of these

Question No. 2 **(Only One Option Correct Type)**

If $\vec{r}\cdot\hat{i}=2\vec{r}\cdot\hat{j}=4\vec{r}\cdot\hat{k}$ and $|\vec{r}|=\sqrt{84}$, then the value of $\vec{r}\cdot\left(2\hat{i}-3\hat{j}+\hat{k}\right)$ may be

- 0
- 2
- 4
- 6

Question No. 3 **(Only One Option Correct Type)**

Let $\mathrm{A}(2,3,-1)$ be a point in space and B be a point on the line $\frac{x-4}{3}=\frac{y-2}{4}=\frac{z-4}{-5}=t$. The vector $\overrightarrow{\mathrm{AB}}$ is perpendicular to the line $\frac{x-3}{3}=\frac{y-8}{1}=\frac{z-3}{1}$. Evaluate $|4\,t|$.

- 5
- 9
- 6
- 4

Question No. 4 **(Only One Option Correct Type)**

Let $A = (0, 4)$ and $B = (2\cos\theta, 2\sin\theta)$, for some $0 < \theta < \frac{\pi}{2}$. Let P divide the line segment AB in the ratio $2 : 3$ internally. The locus of P is

- parabola
- ellipse
- circle
- hyperbola

Question No. 5 **(Only One Option Correct Type)**

Let n be the coefficient of x^6 in $(1 - x^3)^5 (1 + x^2)^4 (1 + x^4)^8$, when expanded in powers of ' x '. The value of $\left[\frac{n}{19}\right]$ is (where [.] denotes greatest integer function)

- 1
- 3
- 2
- 5

Question No. 6 **(Only One Option Correct Type)**

Let $a_n = 16, 4, 1, \ldots\ldots$ be a geometric sequence. The value of $\sum_{n=1}^{\infty} \sqrt[n]{P_n}$, where P_n is the product of the first n terms, is equal to

- 8
- 16
- 32
- 64

Question No. 7 **(Only One Option Correct Type)**

If $\frac{xdy}{dx} - y = \sqrt{x^2 + y^2}$, then

- $x + \sqrt{x^2 + y^2} = cy^2$
- $\sqrt{x^2 + y^2} - y = cx^2$

- $\sqrt{x^2+y^2}+y=cx^2$
- $\sqrt{x^2+y^2}-x=cy^2$

Question No. 8 **(Only One Option Correct Type)**

If $\left(x^2-5x+4\right)\left(y^2+y+1\right)<2y$ for all real y, then x belongs to the interval $(2,b)$, then b can be

- 3
- 4
- 5
- 6

Question No. 9 **(Only One Option Correct Type)**

Three circles each of radius 3 are drawn with centers at $(0,16),(3,0)$ and $(5,8)$. A line of slope m passing through $(3,0)$ is such that the total area of the part of the three circles to one side of the line is equal to the total area of the part of the three circles to the other side of it. Then $\left|\frac{m}{8}\right|$ is equal to

- 1
- 4
- 3
- 2

Question No. 10 **(Only One Option Correct Type)**

If n positive integers are taken at random and multiplied together, the probability that the last digit of the product is 2, 4, 6 or 8, is

- $\frac{4^n+2^n}{5^n}$
- $\frac{4^n\times 2^n}{5^n}$
- $\frac{4^n-2^n}{5^n}$
- None of these

Question No. 11 **(Only One Option Correct Type)**

A stick of length a cm rests against a vertical wall and the horizontal floor. If the foot of the stick slides with a constant velocity of $b\,\frac{\text{cm}}{\text{s}}$, then the magnitude of the velocity of the middle point of the stick when it is equally inclined with the floor and the wall, is

- $\frac{b}{\sqrt{2}}\,\frac{\text{cm}}{\text{s}}$
- $\frac{b}{2}\,\frac{\text{cm}}{\text{s}}$
- $\frac{ab}{2}\,\frac{\text{cm}}{\text{s}}$
- $\frac{a}{\sqrt{2}}\,\frac{\text{cm}}{\text{s}}$

Question No. 12 **(Only One Option Correct Type)**

Find the value of $32 \sin\frac{90^\circ}{17} \cos\frac{180^\circ}{17} \cos\frac{360^\circ}{17} \cos\frac{720^\circ}{17}$.

- 1
- 8
- 4
- 2

Question No. 13 **(Only One Option Correct Type)**

A function ' f ' is defined by $f(z) = (4+i)z^2 + \alpha z + \gamma$ for all complex number z, where α and γ are complex numbers if $f(1)$ and $f(i)$ are both real and the smallest possible values of $|\alpha| + |\gamma|$ is p then determine p^2

- 5
- 1
- 3
- 2

Question No. 14 **(Only One Option Correct Type)**

Let $f : A \to B$ be an one to one function
defined as $f(x) = \frac{\sin^{-1}x + \tan^{-1}x}{\cos^{-1}x + \cot^{-1}x}$ then, the number of solutions of the equation
$f\left(x^3 + 14x^2 + 13x - 5\right) = f\left(1 - x^2 + x^3\right)$ is

- 0
- 1
- 2
- 3

Question No. 15 **(Only One Option Correct Type)**

The number of real values of k for which the lines $\frac{x-k}{4} = \frac{y-1}{2} = \frac{z+1}{1}$ and $\frac{x-(k+1)}{1} = \frac{y}{-1} = \frac{z-1}{2}$ are intersecting is

- 0
- 2
- 1
- Infinite

Question No. 16 **(Only One Option Correct Type)**

Component of vector $\bar{p} = 2\hat{i} + 4\hat{j} - 3\hat{k}$ in the direction perpendicular to the direction of vector $\vec{q} = \hat{i} - \hat{j} + \hat{k}$ is $a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, then the value of $a_1 + a_2 + a_3$ is K, find $3K$

- 17
- 14
- 19
- 11

Question No. 17 **(Only One Option Correct Type)**

Let A & B are two non singular matrices of order 3 such that $A + B = I$ & $A^{-1} + B^{-1} = 2I$, then $|\text{adj}(4AB)|$, is (where $\text{adj}(A)$ is adjoint of matrix A)-

- 4
- 16
- 64
- 128

Question No. 18 **(Only One Option Correct Type)**

The focus of an ellipse is $M(-1,-1)$ and corresponding directrix $x-y+3=0$ and eccentricity is $\frac{1}{2}$. If the centre of the ellipse be $C(p,q)$, then find the value of $(2p-4q)$.

- 4
- 5
- 6
- None

Question No. 19 **(Only One Option Correct Type)**

The number of solutions of the following system of linear homogeneous equations $x-y+z=0, x+2y-z=0$ and $2x+y+3z=0$ is

- 1
- 8
- Countable infinite
- Uncountable

Question No. 20 **(Only One Option Correct Type)**

If $f\left(\frac{2x+3}{3x+5}\right)=x+4, x\neq\frac{-5}{3}, \frac{2}{3}$ and $\int f(x)dx=Ax+B\ln|3x-2|+C$, then $3B-A=$

- $\frac{64}{9}$
- $\frac{-52}{21}$
- $\frac{-10}{3}$
- $\frac{-8}{3}$

Mathematics Numerical (Maximum Marks: 20)

Question No. 1 **(Numerical Type)**

From the string *abacababababcdced*, if 5 letters should be selected, then the number of ways in which this selection can be done is

Question No. 2 **(Numerical Type)**

Find the total number of points of discontinuity and non-differentiability of function $f(x) = \frac{1}{2} - x + \frac{1}{2}[2x] - \frac{1}{2}[1-2x]$ in $[0,1]$. ([x] is greatest integer function)

Question No. 3 **(Numerical Type)**

The coefficient of $a^8b^4c^9d^9$ in $(abc + abd + acd + bcd)^{10}$ is

Question No. 4 **(Numerical Type)**

Let $f(x)$, be a polynomial of degree 3 , such that $f(-1) = 10, f(1) = -6, f(x)$, has a critical point at $x = -1$ and $f'(x)$ has a critical point at $x = 1$. Then $f(x)$, has local minima at $x =$

Question No. 5 **(Numerical Type)**

Find the number of real values of m for which $A \cup B$ has exactly 3 distinct elements, where A and B are following two sets:

$A = \{x : x^2 + (m-1)x - 2(m+1) = 0,\ x \in R\}$

$B = \{x : (m-1)x^2 + mx + 1 = 0,\ x \in R\}$

(ANSWER KEY)

Physics

Q1 [ ]	Q2 [ ]	Q3 [ ]	Q4 [ ]	Q5 [ ]
Q6 [ ]	Q7 [ ]	Q8 [ ]	Q9 [ ]	Q10 [ ]
Q11[ ]	Q12 [ ]	Q13 [ ]	Q14 [ ]	Q15 [ ]
Q16[ ]	Q17 [ ]	Q18 [ ]	Q19 [ ]	Q20 [ ]

Q21 [ ]	Q22 [ ]	Q23 [ ]	Q24 [ ]	Q25 [ ]

Chemistry

Q1 [ ]	Q2 [ ]	Q3 [ ]	Q4 [ ]	Q5 [ ]
Q6 [ ]	Q7 [ ]	Q8 [ ]	Q9 [ ]	Q10 [ ]
Q11[ ]	Q12 [ ]	Q13 [ ]	Q14 [ ]	Q15 [ ]
Q16[ ]	Q17 [ ]	Q18 [ ]	Q19 [ ]	Q20 [ ]

Q21 [ ]	Q22 [ ]	Q23 [ ]	Q24 [ ]	Q25 [ ]

Mathematics

Q1 [ ]	Q2 [ ]	Q3 [ ]	Q4 [ ]	Q5 [ ]
Q6 [ ]	Q7 [ ]	Q8 [ ]	Q9 [ ]	Q10 [ ]
Q11[ ]	Q12 [ ]	Q13 [ ]	Q14 [ ]	Q15 [ ]
Q16[ ]	Q17 [ ]	Q18 [ ]	Q19 [ ]	Q20 [ ]

Q21 [ ]	Q22 [ ]	Q23 [ ]	Q24 [ ]	Q25 [ ]

#dont have it